



NAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE225: Digital Logic Design

Batch: 25th

Date: 20/09/2025

Final Examination

Semester: Spring 2025

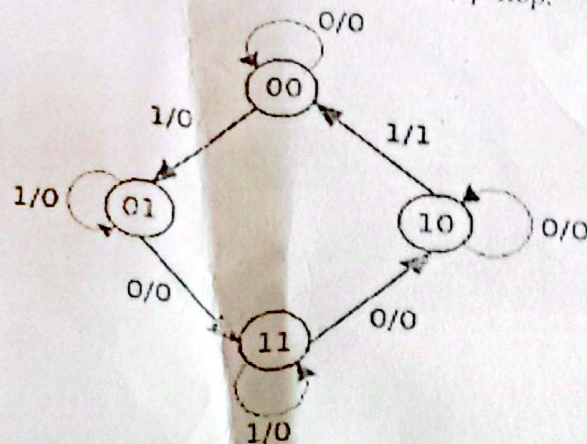
Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

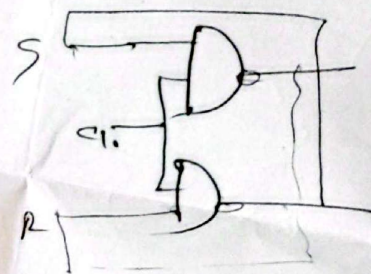
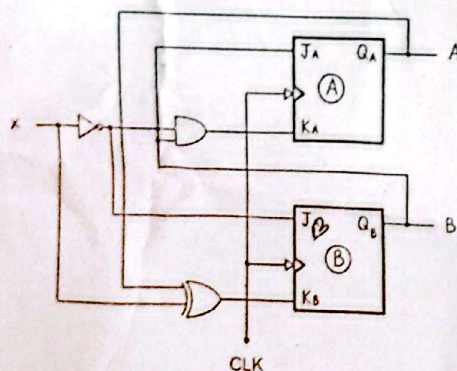
1. (a) Differentiate between combinational circuits and sequential circuits. CLO1 3
(b) Compare and contrast latches and flip-flops, using appropriate diagrams. CLO4 2
(c) Construct an SR latch using NAND and NOR gates, and verify the truth tables for all possible input combinations. CLO1 3
2. (a) Differentiate between the Toggling Problem and the Race-Around Condition. CLO4 3
(b) Design a JK flip-flop and derive its characteristic table and excitation table from its truth table. CLO1 5
3. (a) Mention the five steps involved in a flip-flop conversion. Convert a JK flip-flop into a D flip-flop. CLO1 4
(b) Demonstrate how an SR flip-flop can be converted into a JK flip-flop. CLO1 4
4. (a) Derive the circuit excitation table for the given state diagram to design a clocked sequential circuit using a T flip-flop. CLO5 4



- (b) Perform state reduction on the following state diagram and draw the reduced state diagram. CLO1 4

Present State	Next State		Present Output	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	B	C	0	0
B	D	E	0	0
C	F	G	0	0
D	H	I	0	0
E	J	K	0	0
F	L	M	0	0
G	N	P	0	0
H	A	A	0	0
I	A	A	0	0
J	A	A	0	1
K	A	A	0	0
L	A	A	0	1
M	A	A	0	0
N	A	A	0	0
P	A	A	0	0

5. (a) Explain the differences between Moore and Mealy state machines, CLO1 3
using appropriate diagrams.
- (b) For the given sequential circuit, analyze the possible state CLO1 5
transitions and output sequences and finally draw a state diagram.



6. (a) What are the primary types of counters, and how do they differ from one another? CLO4 3
- (b) Design a 3-bit asynchronous up counter and provide a logical explanation of its operation. CLO5 5
7. (a) How does a PLA differ from a PAL in terms of its architecture and functionality? CLO4 2
- (b) Given the Boolean functions $Y_1 = AB + C$ and $Y_2 = A'C + BC$, design a PAL that implements these functions. CLO5 3
- (c) Design the block diagram of an 8x8 ROM and explain its working. CLO5 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE229: Computer Architecture

Batch: 25th

Date: 17/09/2025

Final Examination

Semester: Spring 2025

Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

- | | | | |
|----|---|------|---|
| 1. | What are Virtual Memory & Cache Memory? | CLO3 | 2 |
| | Explain the Address Translation of a virtual memory with proper diagram. | CLO3 | 3 |
| | Draw the architecture of a virtual memory system with TLB (Translation Lookaside Buffer) in computer system. | CLO1 | 3 |
| 2. | (a) Define the principle of locality of reference. What is meant by the Write-Through and Write-Back protocols? | CLO3 | 3 |
| | (b) Discuss the different types of cache mapping techniques. Mention their respective advantages and disadvantages. | CLO1 | 5 |
| 3. | (a) What are PROM and EPROM? Draw and explain the structure of a static RAM (SRAM) cell. | CLO5 | 3 |
| | (b) Differentiate between hard failure and soft error. Draw and explain the organization of a synchronous DRAM. | CLO5 | 5 |
| 4. | Illustrate the organization of bit cells in a 16×8 memory chip. | CLO4 | 4 |
| | Illustrate the organization of a 1K×1 memory chip. | CLO4 | 4 |
| 5. | (a) What are the possible ways of storing data in memory? Show the memory hierarchy in terms of speed, size and cost. | CLO3 | 3 |
| | (b) Draw the internal organization of a 2M×8 dynamic memory chip. | CLO5 | 5 |
| 6. | (a) What is pipelining? Draw a timing diagram showing the execution of five instructions in a pipelined processor. | CLO5 | 4 |
| | (b) What is hazard in pipelining? Describe the various types of data hazards with appropriate examples. | CLO5 | 4 |
| 7. | (a) Explain control hazards in pipelining. What are the solutions that are commonly applied to minimize them? | CLO5 | 4 |
| | (b) Write short notes on the following: | | |
| | i. Memory Cycle Time | CLO5 | 4 |
| | ii. Memory Address Register | | |
| | iii. Page Table | | |
| | iv. Cache Miss | | |

↓
Code
Address

Some
memory
error

AMDA UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 25th
Date: 15/09/2025

Final Examination

Semester: Spring 2025
Time: 2 Hours
Total Marks : 40

baseband

Answer any five questions:

5 × 8 = 40

	CLOs	Marks
1. (a) Mention the advantages of optical fiber over twisted-pair and coaxial cable.	CLO1	3
(b) How does sky propagation differ from line-of-sight propagation? Explain.	CLO3	5
2. (a) What are the differences between parallel transmission and serial transmission?	CLO4	4
(b) Define the types of Digital-to-Analog conversions.	CLO3	4
3. (a) Explain the three strategies for data rate management in TDM.	CLO5	4
(b) Describe the digital hierarchy used in Data Communication.	CLO1	4
4. (a) Differentiate between T-line and E-line.	CLO1	4
(b) Define FHSS and explain how it achieves bandwidth spreading.	CLO4	4
5. (a) List the categories of unshielded twisted-pair cable.	CLO1	4
(b) How do guided media differ from unguided media? Explain.	CLO1	4
6. (a) Describe the phases in Circuit Switched network.	CLO4	4
(b) List the characteristics of a Virtual-circuit network.	CLO4	4
7. (a) Distinguish between baseband transmission and broadband transmission.	CLO4	4
(b) Compare and contrast a circuit-switched network and a packet-switched network.	CLO4	4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

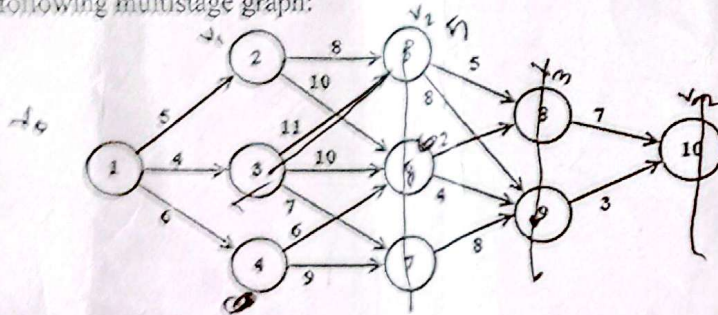
Final Examination
Semester: Spring 2025

Batch: 25th
Date: 08-09-2025

Time: 2 Hours
Total Marks: 40

CLOs	Marks
CLO3	2
CLO5	6

- (a) Why is memorization sometimes preferred over tabulation?
(b) Calculate the minimum cost from node 1 to 10 using dynamic programming for the following multistage graph:



$$\text{Cost}(1,7) = \min(\text{Cost}(1,5) + 7, \text{Cost}(1,6) + 4)$$

- (a) Why is dynamic programming often better than recursion for combinatorial problems?
(b) Consider the following instance of the 0-1 knapsack problem:

Number of items, $n = 4$.

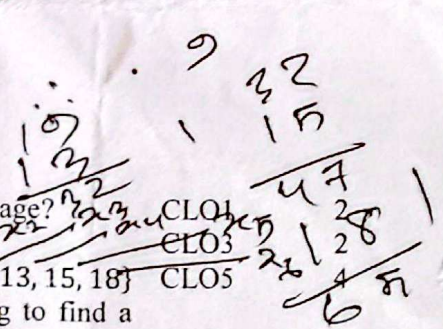
Capacity, $W = 10$,

Profits $\{p_1, p_2, p_3, p_4\} = \{5, 8, 10, 14\}$ and

Weights $\{w_1, w_2, w_3, w_4\} = \{2, 4, 5, 7\}$

Calculate the optimal solution using dynamic programming.

3. (a) What is the difference between BFS and DFS in terms of memory usage?
(b) How does Dijkstra update the distance of neighboring vertices?
(c) Suppose you have 6 elements with the weights, $w[1:6] = \{4, 5, 10, 13, 15, 18\}$ and capacity $m = 28$. Draw the state space tree using backtracking to find a possible subset of w that sum to m .



4. (a) Distinguish between backtracking and recursion.
(b) Find a solution for n -queens problem using backtracking when $n = 4$.

- (a) What happens in Branch and Bound if the bounding function is weak or overestimates too much?
(b) Apply Branch and Bound technique to solve 15-puzzle problem for the particular organization:

1	2	3	
5	6	8	4
9	10	7	11
13	14	15	12



6. (a) Why does the load factor of open addressing always lie between 0 and 1? CLO3 2
(b) Differentiate between Closed Hashing and Open Hashing. CLO3 3
(c) A hash table of length 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below. CLO3 3

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

- i. 46, 42, 34, 52, 23, 33
ii. 34, 42, 23, 52, 33, 46
iii. 46, 34, 42, 23, 52, 33
7. (a) Prove that any decision tree that can sort n elements must have the height $n \log n$. CLO1 3
(b) Prove that the following problem is NP-hard. CLO4 5
Given an undirected graph G , does G contain a simple path that visits all but 17 vertices?

1 → 4 → 6 → 9 → 10



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming

Batch: 25th

Semester: Spring 2025

Time: 2 Hours

Total Marks: 40

Date: 13/09/2025

Final Examination

5 × 8 = 40

Answer any five questions of the following.

CLOs Marks

CLO 1 2

CLO 2 6

- 1(a). How does a subclass execute its superclass constructor?
1(b). Consider the following Java code with main method:

```
class Animal{
    private String name;
    private int age;

    public Animal(String name,int age){
        this.name=name;
        this.age=age;
    }
}

public class InhetanceJava {
    public static void main(String[] args) {
        Bird alex=new Albatross("Alex",39);
        Mammal spot=new Dog("Spot",7);
        Reptile steph=new EasternBrownSnake("Steph",12);
        Fish bruce=new GreatWhiteShark("Bruce",40);
        Arachnid charlotte=new RedBackSpider("Charlotte",1);
        Animal[] animals={alex,spot,steph,bruce,charlotte };
    }
}
```

Write Java code for the necessary classes to support the main method where you cannot create any object of the Animal class. You can add/modify code to the Animal class if required.

- 2(a). What is 'thread' in Java? How does multithreading upgrade the CPU utilization? Explain with an example in brief. CLO 3 3
- 2(b). Write a Java program that uses the **ScheduledExecutorService** interface to schedule tasks for execution at a specified time or with a fixed delay. CLO 2 5
- 3(a). Is the given statement accurate: 'A class with an abstract method must CLO 1 3

- also be declared as abstract? Justify your response.
- 3(b). Consider the following code segment:

<pre>interface x1 { void f1(); static void f2() {} default void f3() {} } interface x2 { void f4(); private void f5() {} }</pre>	<pre>abstract class A implements x1 { final void f6() {} abstract void f7(); } public class B extends A implements x2 { // your code }</pre>
---	---

Demonstrate successful compilation of class B by writing the minimum code. You cannot define class B as abstract.

- 4(a). Consider the string "MadAm". Write a program in java to find if the given string is palindrome or not. Here, the string case is insensitive.
- 4(b). Write a Java program to get the character at the given index within the string.

Sample input/output:

Original String = Java Exercises!

The character at position 0 is J

The character at position 10 is i

- 5(a). Explain 'Autoboxing' and 'Unboxing' with examples.
- 5(b). What is the 'Wrapper' class in Java? Briefly explain the relation between Wrapper class and ArrayList with an example.

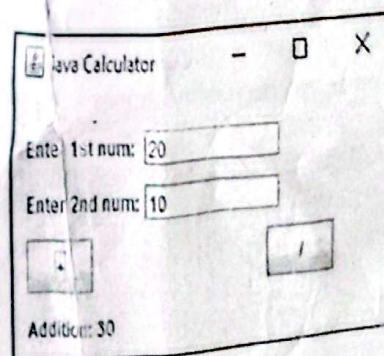
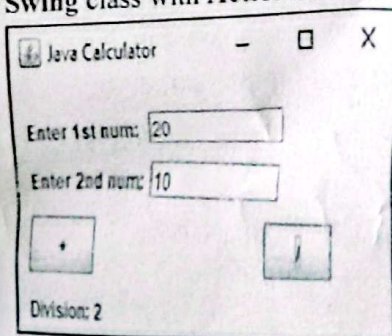
- 6(a). Define Abstraction and Encapsulation in Object Oriented Programming.

- 6(b). Write four ActionListener built-in method names.

- 6(c). Explain Java Abstraction method and class with examples.

- 7(a). What is the use of an *import* declaration in a Java program?

- 7(b). Write a Java program to display the following interface using AWT, Swing class with ActionListener.





HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

ENS 221: Environmental Science

Final Examination

Date: 10/09/2025

Time: 2 Hours

Semester: Spring 2025

Total Marks: 40

Part A (Answer any Two)

CLO Marks

- | | | | |
|---|---|------|-----|
| 1 | (a) Define environment? Environmental Science is a multi-disciplinary approach. Explain. | CLO1 | 6 |
| | (b) Differentiate between Environmental Scientist and Environmentalist? Provide examples. | CLO1 | 4 |
| 2 | (a) How do smart cities contribute to reducing environmental impacts while enhancing urban living? Provide examples of technologies used to achieve this. | CLO3 | 5 |
| | (b) Discuss the challenges faced by cities in developing countries when implementing smart city technologies aimed at sustainability. | CLO3 | 5 |
| 3 | (a) Define disaster with formula and explain its key components. | CLO2 | 2.5 |
| | (b) How is disaster management in Bangladesh different from the global approach to disaster management? | CLO3 | 2.5 |
| | (c) Discuss the major flood that have affected Bangladesh. How does the geographical location of Bangladesh contribute to its vulnerability to cyclones and flooding? | CLO3 | 5 |

Part B (Answer any Two)

- | | | | |
|---|--|------|---|
| 4 | (a) Define Green Tourism. Identify the drawbacks of developing eco-tourism in the context of Bangladesh. | CLO4 | 5 |
| | (b) Apply the prime indicators of eco-tourism for ensuring sustainability. | CLO4 | 5 |
| 5 | (a) Compare between adaptation and mitigation. | CLO3 | 3 |
| | (b) Write Six Pillars of Bangladesh Climate Change Strategy Action Plan and explain the first three pillar. | CLO3 | 7 |
| 6 | (a) How you can delineate carbon trading and low carbon development. | CLO1 | 3 |
| | (b) Calculate how much a Jackfruit tree sequesters carbon in a year. A 15-year-old Jackfruit tree would probably grow about 22 feet tall with a trunk about 12 inches in diameter. | CLO4 | 7 |